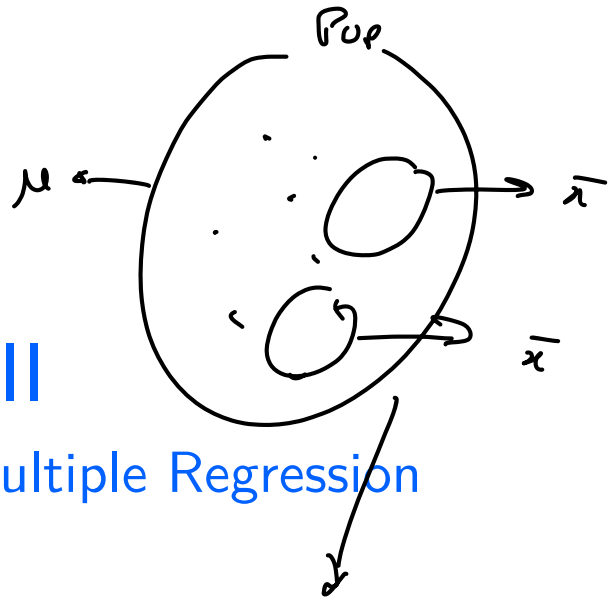


6 - Linear Regression II

Inference for Regression and Motivation for Multiple Regression

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Agenda

Today we move from **estimating a regression line** to asking how much we can trust and interpret it.

1. From an estimated slope to uncertainty
2. Inference for regression coefficients
   - Confidence intervals
   - Hypothesis tests and p-values
   - Statistical significance vs. substantive meaning
3. Why simple regression can mislead
   - Intuition for multiple regression

Main question: How uncertain is an estimated relationship, and when might a simple regression be misleading?

Recommended reading: Wooldridge, Chapter 2.5 and Chapter 3.1

Investigating the Drivers of Laptop Price

Question: How is laptop price related to RAM?

You estimate the simple regression:

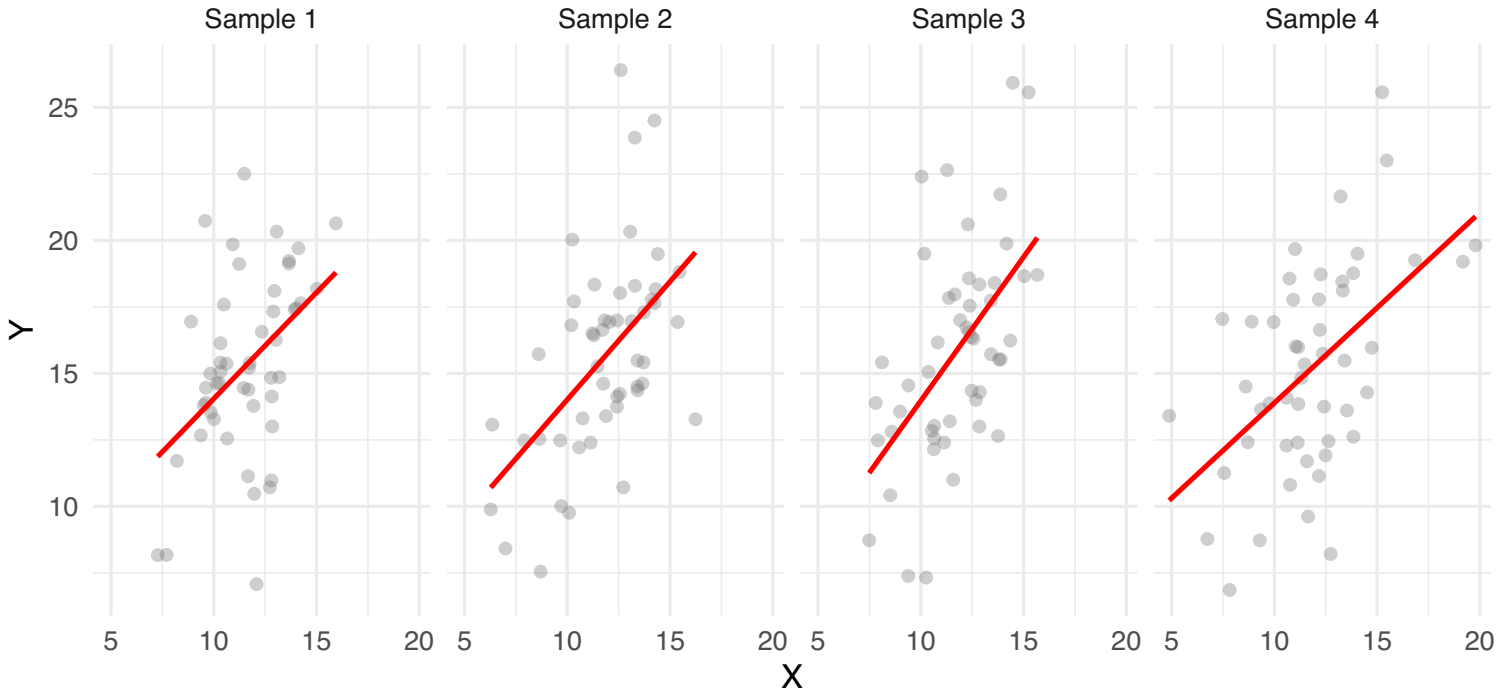
$$\widehat{Price}_i = 276 + 101.8 \cdot RAM_i$$

Now you want to know:

- Is the estimated slope **precise or noisy**?
- Is the relationship **statistically distinguishable from zero**?
- Is RAM itself the driver, or is RAM bundled with **other premium features**?

Recall: Different Samples, Different Lines

If we took another random sample from the same population, OLS would generally give us a different line.

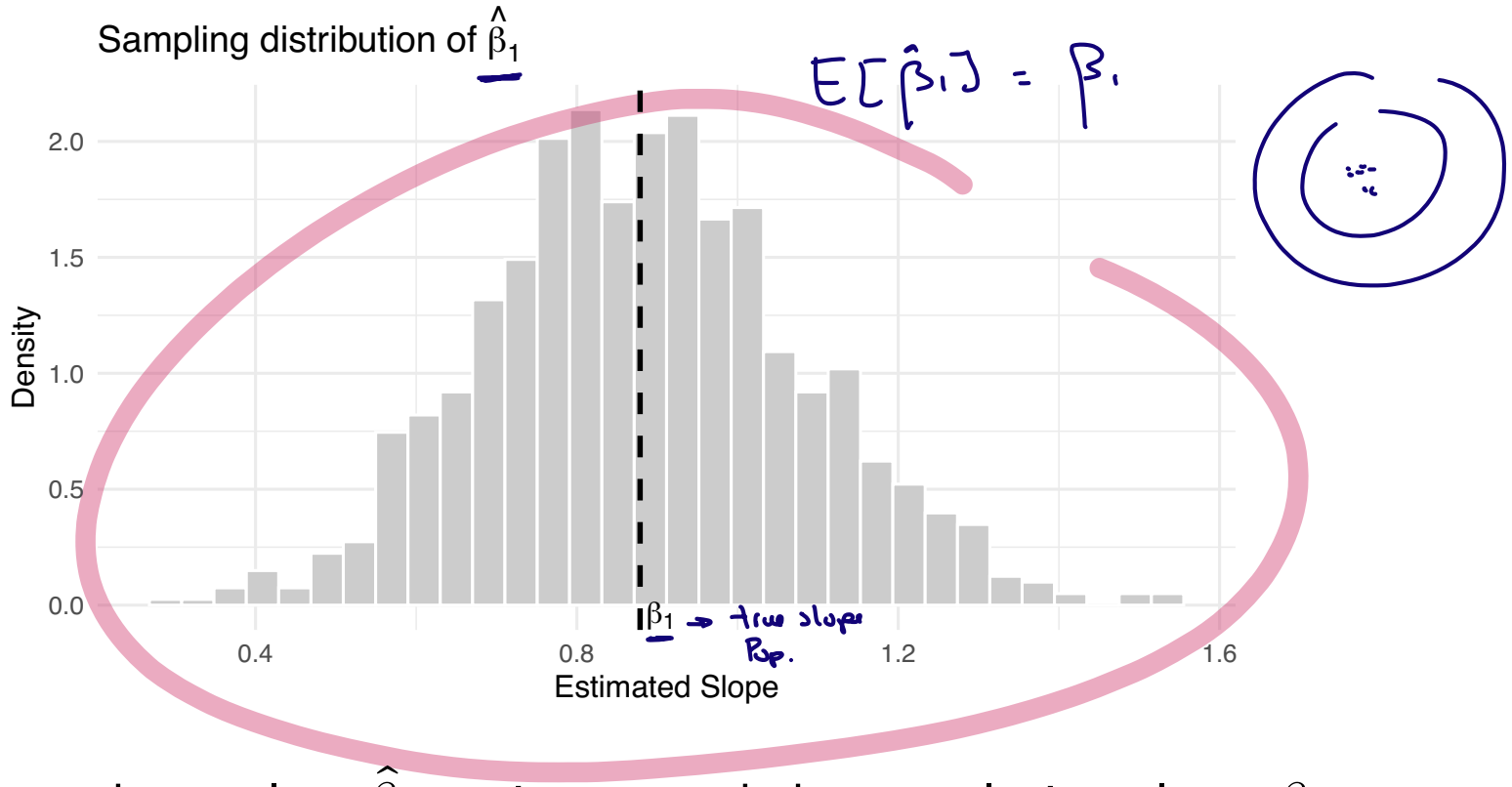


The population slope β_1 is fixed. The estimate $\hat{\beta}_1$ changes from sample to sample.

1. Same Inferential Machinery, New Parameter

The Sampling Distribution of $\hat{\beta}_1$

In this simulated example, we can repeatedly draw samples and estimate a slope each time.



Across repeated samples, $\hat{\beta}_1$ varies around the population slope β_1.

From Means to Regression Coefficients

The object changed. The inferential logic did not.

For a population mean:

Pop mean → μ ← $\bar{Y}$ ← Sample mean

$$CI = \bar{Y} \pm t^* SE(\bar{Y})$$

Estimate ± Critical value × SE of Estimate

$$T = \frac{\bar{Y} - \mu_0}{SE(\bar{Y})}$$ ← null value

For a regression slope:

Pop. (true) slope → β_1 ← $\hat{\beta}_1$ ← Estimate

$$CI = \hat{\beta}_1 \pm t^* SE(\hat{\beta}_1)$$

$$T = \frac{\hat{\beta}_1 - \beta_1^0}{SE(\hat{\beta}_1)}$$ ← null value

Most commonly, we test:

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_a : \beta_1 \neq 0$$

Price = $\beta_0 + \beta_1$ RAM $+ u$

Standard Error of the Slope

Under the classical simple-regression assumptions:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2} = \frac{\hat{\sigma}^2}{(n-1)s_x^2}$$

and in practice we estimate it using the residual variation from the fitted model.

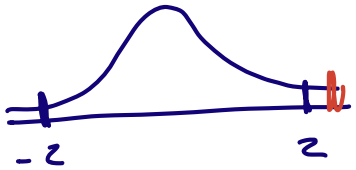
Understanding the formula gives us the intuition:

- more noise around the line $\rightarrow$ less precision;
- more observations $\rightarrow$ more precision;
- more variation in x $\rightarrow$ more precision.

2. Reading Regression Output

Regression Output for the Laptop Model

We estimate:



$$Price = \beta_0 + \beta_1 RAM + u$$

Table 1: Simple regression: Laptop price on RAM

Term	Estimate	Std. Error	t value	p-value
Intercept	276.03 ($\hat{\beta}_0$)	25.54	10.81	$< .001$
RAM	101.76 ($\hat{\beta}_1$)	2.59	39.29	$< .001$

95% CI for β_1 : $101.76 \pm \frac{1.96}{2} \times 2.59$

$$R^2 = 0.55 \qquad SER = €471$$

$H_0: \beta_1 \leq 0$

$H_a: \beta_1 > 0$

$H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

$$\tau = \frac{101.76 - 0}{2.59} = 39.29$$

What Does the RAM Row Tell Us?

For RAM:

$$\hat{\beta}_1 = 101.8 \qquad SE(\hat{\beta}_1) = 2.6$$

A 95% confidence interval is approximately:

$$[96.7,\ 106.8]$$

Interpretation:

We are 95% confident that, on average, one additional GB of RAM is associated with a price increase between 96.7 *and* 106.8 *euros.*

Testing Whether the Slope Is Zero

The usual test asks whether there is evidence that the population slope is different from zero:

$$H_0 : \beta_1 = 0 \qquad\qquad H_a : \beta_1 \neq 0$$

The test statistic is:

$$T = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{101.8}{2.6} = 39.29$$

The p-value is $< \mathbf{.001}$. < 0.05 → Reject H_0 that $\beta_1 = 0$ statistically sig. result

This gives strong evidence that the population slope is not zero in this simple regression.

Statistical Evidence Is Not the Same as Business Meaning

A statistically significant slope answers one question:

> *Is the estimated relationship large relative to its sampling uncertainty?*

It does **not** automatically answer:

- Is the effect economically important?
- Is the relationship causal?
- Is RAM the actual driver of price?
- Would the relationship hold for a new set of laptops?

For business analytics, always interpret:

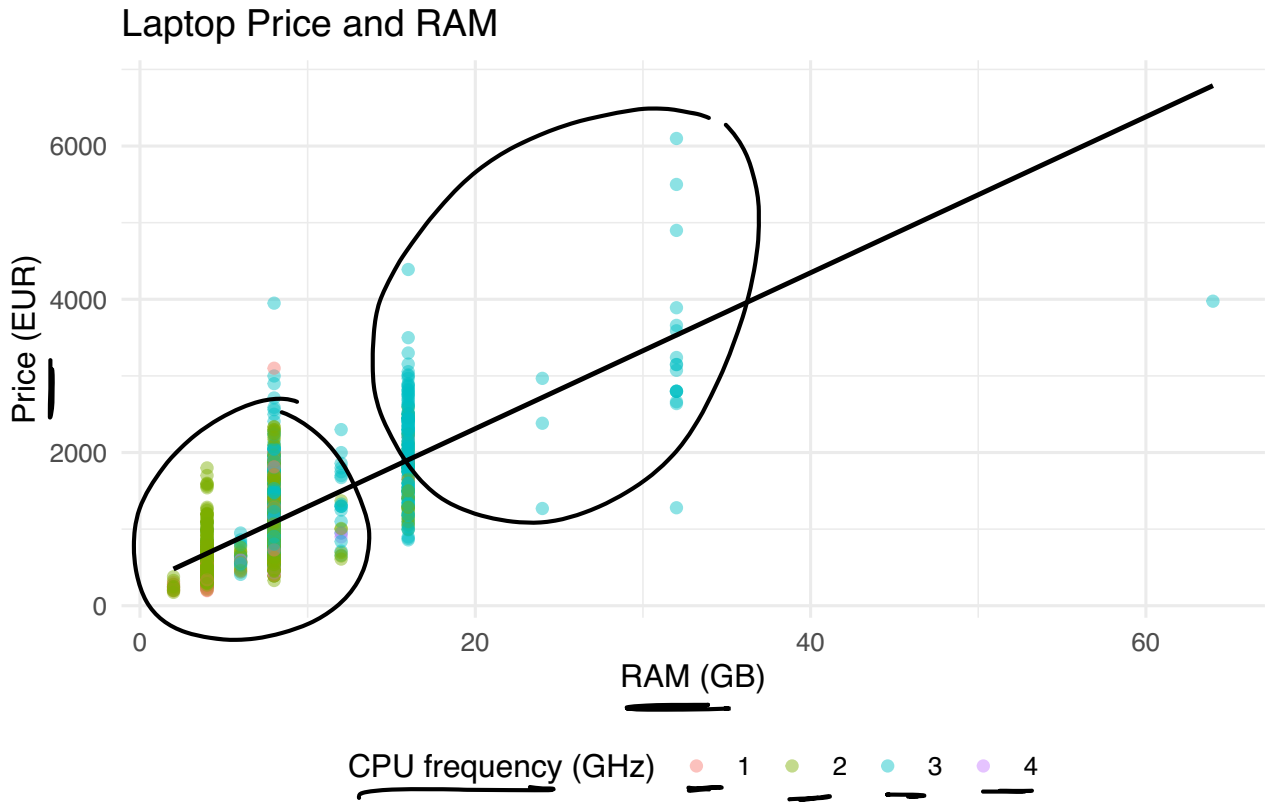
$$\text{estimate} + \text{uncertainty} + \text{substantive context}$$

3. Why Simple Regression Can Mislead

What Else Moves with RAM?

Simple regression compares laptops with different RAM values.

But higher-RAM laptops may also differ in other ways.



In this dataset, $Corr(RAM, CPU\ \ frequency) = 0.37$.

A Simple Regression May Mix Several Differences

The simple regression slope compares laptops with different RAM values.

But those laptops may also differ in:

- processor quality;
- storage;
- screen size or resolution;
- brand;
- product segment.

So the simple slope may combine:

$$\text{RAM difference} + \text{other differences correlated with RAM}$$

This is why an association can be statistically strong but still incomplete or misleading.

What Happens When We Add One Control?

Now compare:

- Model 1: $Price = \beta_0 + \beta_1 RAM + u$
- Model 2: $Price = \beta_0 + \beta_1 RAM + \beta_2 CPU\,freq + u$

Table 2: How the RAM estimate changes when CPU frequency is included

Quantity	SLR: Price ~ RAM	MLR: Price ~ RAM + CPU_freq
RAM coefficient	€101.8	€92.6
Std. Error of RAM coefficient	€2.6	€2.7
p-value for RAM	$< .001$	$< .001$
R2	0.548	0.577
SER	471	456

The First Intuition of Multiple Regression

Multiple regression lets us make a more like-for-like comparison by holding other included variables fixed.

For example:

$$\widehat{Price} = -230 + 93 \cdot RAM + 253 \cdot CPUfreq$$

Here, $\hat{\beta}_1$ describes how price changes with RAM while **holding CPU frequency fixed**.

> *Holding CPU frequency fixed, laptops with 1 additional GB of RAM are estimated to cost, on average, about € 93 more.*

4. Closing

Key Takeaways

1. $\hat{\beta}_1$ is an **estimate** of the population slope β_1.
2. Because random samples differ, $\hat{\beta}_1$ has a **sampling distribution** and a **standard error**.
3. Confidence intervals and hypothesis tests for regression coefficients use the **same logic** as before.
4. A statistically significant coefficient is not automatically **causal** or **business-important**.
5. Simple regression may be misleading when the explanatory variable is correlated with other relevant factors.

Next: multiple regression and the meaning of **ceteris paribus**.

Appendix / Reference

One-Sided Alternatives?

Most regression software reports the **two-sided** p-value for regression coefficients. For a one-sided alternative, use the sign of the t-statistic:

- If $H_a : \beta_j > 0$ and $t > 0$ or $H_a : \beta_j < 0$ and $t < 0$:

$$p_{\text{one-sided}} = \frac{p_{\text{two-sided}}}{2}$$

- If the estimate goes in the **opposite direction**, the one-sided p-value is large:

$$p_{\text{one-sided}} = 1 - \frac{p_{\text{two-sided}}}{2}$$

In R, you can also calculate it directly from the t-statistic:

```
# Ha: beta_j > 0
pt(summary(m2)$coefficients["Ram", "t value"], df = summary(m2)$df[2], lower.tail = FALSE)
```

```
[1] 7.181264e-184
```

```
# Ha: beta_j < 0
pt(summary(m2)$coefficients["Ram", "t value"], df = summary(m2)$df[2], lower.tail = TRUE)
```

```
[1] 1
```

Why Do Coefficients Change When We Add a Variable?

Consider a regression with two explanatory variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

For two regressors, the OLS slopes can be written as:

$$\hat{\beta}_1 = \frac{s_y}{s_{x_1}} \frac{r_{x_1 y} - r_{x_1 x_2} r_{x_2 y}}{1 - r^2_{x_1 x_2}}$$

$$\hat{\beta}_2 = \frac{s_y}{s_{x_2}} \frac{r_{x_2 y} - r_{x_1 x_2} r_{x_1 y}}{1 - r^2_{x_1 x_2}}$$

The coefficient on x_1 therefore depends not only on its relationship with y, but also on how x_1 is related to x_2.

A useful special case: if x_1 and x_2 are uncorrelated,

$$r_{x_1 x_2} = 0 \quad \Longrightarrow \quad \hat{\beta}_1 = r_{x_1 y} \frac{s_y}{s_{x_1}}$$

which is exactly the **simple-regression slope** of y on x_1.

Recommended Readings

- Wooldridge, *Introductory Econometrics: A Modern Approach*, 7th ed., Chapter 2
  - Section 2.5: Expected Values and Variances of OLS Estimators
  - Section 3.1: Motivation for Multiple Regression